1 new onepageer

Let S be a semigroup, $m(S) > 2$ and $V = \{z \in \mathbb{Z}_{\geq 0} \mid z \neq 0 \mod m, z \neq \mu \mod m\}$

2 Ascent-Flip

Let $X = (\mathbb{Z}_{>0} \operatorname{Atom}(S)) \cap V \cap \operatorname{Ap}(S)$ be the set of all elements of V that are integer multiples of an atom and Apery, i.e. $X = \{z \in V \mid \exists a \operatorname{Atom}(S), a|z, z - m \notin S\}$
Remark: If ka is Apery for an atom a, then all elements $a, 2a, \ldots, ka$ are Apery.
By construction, $f + m$ and m cannot be Elements of X, because they are not in V
X is not empty, because $(\mathrm{A}(S) \setminus \{m, f + m\}) \subseteq V$ and atoms $\neq m$ are Apery.
Let $w = \max(X)$. Then we define

$$a_{\uparrow}(S) = \max\{a \in \operatorname{Atom}(S) \mid a|w\}$$

This always exists, is $< f + m$, and we let $A(S) = S \setminus \{a_{\uparrow}\}$

3 Descent-Flip

Let $Y = \operatorname{Ap}(S) \cap V$ be the set of all Apery-Elements in V
$Y = \{w_1, \ldots, w_{\mu-1}, w_{\mu+1}, \ldots, w_{m-1}\} = \operatorname{Ap}(S) \setminus \{0, f + m\}$.
Y is not empty, because we assumed $m > 2$
Let $w = \max(Y)$. Then $w - m$ is the largest gap in V (TODO: $w - m$ is simply $\max((S) \setminus \{f\})$ and we define

$$L_{\downarrow} = w - m > 0$$

and $D(S) = S \cup \{L_{\downarrow}\}$

4 Ascent-Total

If $f + m = a \in \operatorname{Atom}(S)$ let $a_{\uparrow} = f + m$ and define

$$A_t(S) = \langle \operatorname{Atom}(S) \setminus \{a_{\uparrow}\} \rangle$$

(this is well defined because $m > 2$). If $f + m$ is decomposable, define

$$A_t = \langle S \setminus \{f + m\} \rangle$$

5 Descent-Total

Let $L = \max(V \cap G(S))$ be the largest gap in V and let $l \geq 0$ be the natural number such that $lm < L < (l + 1)m$. Since $L < f$ and if $l > 0$ we have $\mu + (l - 1)m$ is a gap (since otherwise $f < L$), we must have $\mu + lm \leq f$ and so we define:

$$L_{\downarrow} = \mu + lm$$

and $D(S) := S \cup \{L_{\downarrow}\}$
comment: if $l = 0$ in the above $m(D(S)) = \mu(S)$ and $\mu(D(S)) < \mu(S) < m$

6 new descent/ascent

Let S be a numerical semigroup with the usual f, m, g, σ, r and Apéry set $\{w_0, \ldots, w_{m-1}\}$ indexed by residue. The four operators below each return a new numerical semigroup.

6.1 DescentFlip

Walk strata $l = 0, 1, \ldots, \text{level} - 1$. At the first stratum that contains an Apéry element w of S with $w \neq f + m$, pick the largest such w and add $w - m$ as a generator (this is a gap of S, in the same residue class as w). No-op if $f < m$ or no such w exists.

```
/// `a_↑ = max{a  Atom(S) : a  w}`. Return `S \ {a_↑}`, i.e. the
/// maximal subsemigroup of `S` not containing `a_↑` ([`Self::toggle`]).
///
/// For `m > 2`, `X` is non-empty: every atom in `Atom(S) \ {m, f+m}`
/// is itself in `X` (atoms are Apéry, in `V` by `w_  = f+m`), and
/// that set is non-empty (else `Atom(S) = {m, f+m}` would force
/// embedding dimension 2 and hence `m = 2` via the 2-generator
/// Frobenius identity).
///
/// Returns `self` when `m < 3`.
#[must_use]
pub fn ascent_flip(&self) -> Self {
    if self.m==1 {return compute(&[2,3])}
    if self.m==2 {return compute(&[2,self.ae+2])}
    // X = {w_r : r  V}  {multiples of some atom}.
    let Some(w) = (1..self.m)
        .filter(|&r| r != self.mu)
        .map(|r| self.apery_set[r])
        .filter(|&w| self.gen_set.iter().any(|&a| w.is_multiple_of(a)))
```

6.2 AscentFlip

Reverse of DescentFlip. Walk strata $l = 0, 1, \ldots, \text{level} - 1$ and $k = 1, \ldots, \text{level}$. Pick the largest atom a such that ka is an Apéry element of S landing in stratum l and $ka \neq f + m$. Remove a from the generators (equivalent to toggling a).

6.3 DescentTotal

Walk strata $l = 1, 2, \ldots$ and stop at the first l at which the open interval $(f - (l+1)m,\ f - lm)$ is *not* fully in S. Candidate new generator: $x := f - (l-1)m$ (a gap of residue μ). Commit the candidate only if applying AscentTotal to the result returns S; otherwise no-op. The safety gate rejects two situations: (i) μ shifts because some w_i with $i \neq \mu$ exceeds x; (ii) a residue-μ generator $g > x$ of S becomes reducible as $g = x + km$ and is dropped from the result.

```
        return self.clone();
    };
    let l = big_l / self.m;
    let target = self.mu + l * self.m;
    self.toggle(target)
}

/// Descent (flip): walks the strata `(f - (l+1)m, f - l m)` for
/// `l = 0, 1, …, level - 1` and, at the first stratum that contains
/// an Apéry element of `S`, picks the largest such Apéry `w` and
/// adds `w - m` as a new generator. The added value is always a gap
/// (the predecessor of `w` in the same residue class).
///
/// **Never adds `f`**: the `l = 0` stratum filter is `w  (f, f + m)`
/// strictly, which excludes `w = f + m`. When the only Apéry above
/// `f` is `f + m` and no in-stratum match is found at any `l`, the
/// function returns `self`.
///
/// Returns `self` when `f < m` (`S =  `) or when `m < 3` (for `m = 2`
/// the flip move is structurally vacuous; use [`Self::descent_total`]
/// instead).
///
/// Invertibility is **not** enforced here. `ascent_flip` is the
/// structural reverse but their composition is the identity only
/// under conditions we do not check at call time; see
/// [`Self::ascent_flip`].
#[must_use]
pub fn descent_flip(&self) -> Self {
    if self.m==1 {return self.clone()}
    if self.m==2 && self.ae>2 {return compute(&[2,self.ae-2])}
    let mut newgen = self.gen_set.clone();
```

6.4 AscentTotal

Destroy the atom $f+m$ when it lies in the generator set, i.e. return $\langle \mathrm{Atom}(S) \setminus \{f+m\}\rangle$. No-op when $f+m \notin \mathrm{Atom}(S)$.

```
pub fn ascent_total(&self) -> Self {
    if self.m==1 {return compute(&[2,3])}
    if self.m==2 {return compute(&[2,self.ae+2])}
    let ae = self.f + self.m;
    if self.gen_set.contains(&ae) {
        self.destroy_atom(ae)
    } else {
        self.toggle(self.f + self.m)
```

6.5 Effect on (g, σ, r) and on e

	DescentFlip	DescentTotal	AscentFlip	AscentTotal
g'	$g-1$	$g-l$	$g+1$	$g+\rho$
σ'	$\sigma+1$	$\sigma+l$	$\sigma-1$	$\sigma-\rho$
r'	$r-2$	$r-2l$	$r+2$	$r+\rho(m-2)$
e' if $f+m \in \mathrm{Atom}(S)$	?	?	?	?
e' if $f+m \notin \mathrm{Atom}(S)$	?	?	?	?

The (g, σ, r) entries follow from $r+\sigma = g$ and the fact that DescentFlip and AscentFlip each flip a single reflected-gap pair. The total operations bundle l (resp. ρ) flips into a single step.
Open question on $e' = e \pm \delta$ when $f+m$ is or is not an atom: e.g. DescentTotal can pull a residue-μ atom down to the new $f+m$ position, making the old $f+m$ reducible ($\sim = x + km$), so δ can be negative; the count depends on how many residue-μ atoms lay above the new x. AscentTotal removes one atom by construction but may make several previously- expressible elements newly minimal. The exact counts are left as follow-up work and tagged ”?” in the table.

7 Setup

Let S be a numerical semigroup, with f, m, g, σ as usual (Frobenius, multiplicity, genus, number of sporadic elements). Write $S_+ := S \setminus \{0\}$.

- $\mathrm{Ap}(S) := \{s \in S \mid s - m \notin S\}$, the *Apéry set* (one representative per residue class mod m). Index it by residue: $\mathrm{Ap}(S) = \{w_0, \ldots, w_{m-1}\}$ with $w_i \equiv i \pmod{m}$. In particular $w_0 = 0$.
- $\mathrm{RG}(S) := \{L \in \{1, \ldots, f-1\} \mid L \notin S,\ f - L \notin S\}$, the *reflected gaps.*
- a is an *atom* of S iff $a \in S_+ \setminus (S_+ + S_+)$. Equivalently, a lies in the unique minimal generating set $\mathrm{Atom}(S)$ of S.
- $\mu \in \{1, \ldots, m-1\}$ denotes $\mu := f \bmod m$; equivalently $f + m = w_\mu$ (the largest Apéry element).
- For each $i \in \{1, \ldots, m-1\}$, set
$$r_i := \frac{w_i + w_{\mu-i} - w_\mu}{m},$$
where the index $\mu - i$ is taken mod m. Geometrically: $w_i + w_{\mu-i}$ sits exactly r_i steps of m above w_μ on the residue-μ ladder, so the closed interval $[w_\mu, w_i + w_{\mu-i}]$ in that ladder contains $r_i + 1$ rungs (the rung $w_\mu - m = f$ being a gap and the rest of the rungs above w_μ being elements of S). r_i counts the reflected gaps in residue class i (mod m).
- $V(S, l) := \{f - lm + 1,\ f - (l-1)m + 2, \ldots, f - 1\}$, the l-Stratum, for $l = 1$ this is the *interval below* f (always $m - 1$ consecutive integers).

- $\rho(S) := \min_{i\in\{1,\dots,m-1\}\setminus\{\mu\}} r_i$. The class $i = \mu$ is excluded because $r_\mu = 0$ trivially (every $x \equiv \mu$ has $f - x \equiv 0 \pmod{m}$, hence $f - x \in S$).
- Total reflected-gap count: $r = \sum_{i=1}^{m-1} r_i = \#\mathrm{RG}(S)$ (the term r_0 would equal 0 anyway).
- Trivial identities: $r + \sigma = g$, $f + 1 + r = 2g$, and $t - 1 \le r$. The first follows from closure: no pair $\{x, f - x\}$ can lie inside S_+ (else $f \in S$), and similarly $f/2$ is always a gap when f is even; pairing up $\{0, 1, \dots, f\}$ under $x \leftrightarrow f - x$ then gives $r + \sigma = g$. The second is $g + \sigma = f + 1$ combined with the first. The third holds because each pseudo-Frobenius $z \neq f$ is a reflected gap ($z + S_+ \subseteq S$ would imply $f \in S$ if $f - z \in S_+$).

8 UpDown

Fix $m > 2$. We define four families of one-step moves, two *descents* ($S \mapsto S^{\downarrow}$, adding one element) and two *ascents* ($S \mapsto S^{\uparrow}$, removing one atom). Primes denote the new counts in $S^{\downarrow}$ resp. $S^{\uparrow}$. The descents are exactly the inverses of the ascents ($\mathcal{A}_{flip}$ inverts $\mathcal{D}_{flip}$, with $w_i \in V(S)$ playing the role of $w_i - m$; $\mathcal{A}_{total}$ inverts $\mathcal{D}_{total}$).

1. $\mathcal{D}_{flip}(i)$, with $i \in \{1, \dots, m-1\} \setminus \{\mu\}$ and $f < w_i < f + m$:

$$S^{\downarrow} := S \cup \{w_i - m\}, \qquad g' = g - 1,\ \sigma' = \sigma + 1,\ r' = r - 2.$$

2. $\mathcal{D}_{total}(\mu)$, requiring $V(S) \subseteq S$:

$$S^{\downarrow} := S \cup \{w_\mu - m\} = S \cup \{f\}, \qquad g' = g - 1,\ \sigma' = \sigma - (m-1),\ r' = r + (m-2).$$

3. $\mathcal{A}_{flip}(i)$, with $i \in \{1, \dots, m-1\} \setminus \{\mu\}$, $ka = w_i$ and Apery–Element for an atom a of S and $k > 0$, and $w_i \in V(S, l)$ (going down on l if necessary):

$$S^{\uparrow} := S \setminus \{a\}, \qquad g' = g + 1,\ \sigma' = \sigma - 1,\ r' = r + 2.$$

4. $\mathcal{A}_{total}(\mu)$, requiring $w_\mu = f + m$ to be an atom of S (equivalently $\rho(S) > 0$); set $\rho := \rho(S)$:

$$S^{\uparrow} := \big\langle A(S) \setminus \{w_\mu\} \big\rangle, \qquad g' = g + \rho,\ \sigma' = \sigma + \rho(m-1),\ r' = r - \rho(m-2).$$

   Note: $S^{\uparrow}$ is *not* the set $S \setminus \{w_\mu\}$. The latter happens to be closed under + already (when $\rho > 0$) but only drops one element; the structural ascent removes $w_\mu, w_\mu + m, \dots, w_\mu + (\rho - 1)m$ in one shot, since those rungs of the residue-μ ladder were reachable in S only via w_μ. The first rung of class μ that survives is $w_\mu + \rho m = w_i + w_{\mu - i}$ for some $i \in \{1, \dots, m-1\} \setminus \{\mu\}$.

All four claims are proved by tracking the Apéry set under the move. Aside: the Wilf conjecture in the form $(e-2)\sigma \ge r$ (which is equivalent to the standard $e\sigma \ge f + 1$ via $r + \sigma = g$ and $g + \sigma = f + 1$) constrains σ and r in tandem. We know the change of both in every case, so working out the change of e would close the loop.

9 Correspondence

We now zoom in on the total cases. Fix $m > 2$ throughout this section.

Equivalence: w_μ atom $\Leftrightarrow \rho(S) > 0$. If $\rho > 0$ then $w_i + w_{\mu-i} = w_\mu + r_i m > w_\mu$ for every $i \in \{1, \dots, m-1\} \setminus \{\mu\}$, so w_μ is not a sum of two positive Apéry elements, hence (since every element of S_+ is $w_j + km$ for some j and $k \geq 0$) not a sum of two elements of S_+ at all; that is, w_μ is an atom. Conversely, if some $r_i = 0$ then $w_i + w_{\mu-i} = w_\mu$ with both summands in S_+, so w_μ is decomposable.

When can $\mathcal{D}_{total}$ and $\mathcal{A}_{total}$ undo each other? Suppose we start at S with $\rho(S) = 0$, so $w_\mu(S)$ is not an atom and we cannot apply $\mathcal{A}_{total}$ to S directly. Apply $\mathcal{D}_{total}$ instead (assuming $V(S) \subseteq S$): we obtain $S^{\downarrow} := S \cup \{f\}$, whose Frobenius drops and whose new w_μ is the atom f (a fresh minimal generator). If the new $\rho(S^{\downarrow}) = 1$, then applying $\mathcal{A}_{total}$ to $S^{\downarrow}$ removes exactly one element — the new w_μ, namely the original f — and returns us to S. This is the one situation in which a single descent followed by a single ascent is a round trip:

$$S_{\rho=0} \xrightarrow{\mathcal{D}_{total}} D(S)_{\rho=1} \xrightarrow{\mathcal{A}_{total}} S.$$

For $\rho(D(S)) > 1$ the round trip is no longer one-step in the ascent: $\mathcal{A}_{total}$ deletes ρ elements at once, so to return to S we would need ρ separate descents (each peeling off one element). Concretely, undoing the ascent's $(g + \rho,\ \sigma + \rho(m-1),\ r - \rho(m-2))$ by ρ applications of $\mathcal{D}_{total}$ gives a net change of

$$g + \rho - \rho \cdot 1 = g, \qquad \sigma + \rho(m-1) - \rho(m-1) = \sigma, \qquad r - \rho(m-2) + \rho(m-2) = r,$$

as required. But each individual descent step demands $V \subseteq S$ at that point, which may fail before the round trip completes; see the next subsection.

The map D. Let $\mathbb{S}_\mu^{\mathrm{old}}$ be the set of numerical semigroups S with $m(S) = m$ and $\mu(S) = \mu$, and satisfying $f(S) > m$ and $V(S) \subseteq S$. Define

$$D : \mathbb{S}_\mu^{\mathrm{old}} \longrightarrow \{\text{numerical semigroups with } m(S) = m\}, \qquad D(S) := S \cup \{f(S)\}.$$

(We do *not* claim $D(S)$ has Frobenius congruent to μ mod m; in general $\mu(D(S)) \neq \mu(S)$. The subscript "μ" refers to the residue of f in S, not in $D(S)$.) The following properties are useful and are proved by tracking the Apéry set through the descent:

- $\rho(D(S)) = \rho(S) + 1$, where on the left ρ is computed with respect to $\mu(D(S))$.
- In residue class i relative to the new $\mu(D(S))$, $r_i(D(S)) = r_i(S) + 1$.
- $\rho(D(S)) > 0$, so $w_\mu(D(S)) = f(S) + m$ is an atom of $D(S)$, i.e. $D(S)$ is eligible for $\mathcal{A}_{total}$.

What remains to control along an iterated descent is whether $V(D(S)) \subseteq D(S)$ — the condition needed to apply $\mathcal{D}_{total}$ a second time.

9.1 One-step bijection

The only way a single $(\mathcal{D}_{total}, \mathcal{A}_{total})$ pair is a round trip is

$$S_{\rho=0} \longleftrightarrow D(S)_{\rho=1},$$

i.e. when $w_\mu(S)$ is *not* an atom of S but $D(S)$ has $\rho = 1$. Only then is $D(S) \in \mathcal{A}_{total}(\mu(D(S)))$ and the ascent's removal-of-ρ-elements coincides with the descent's addition of f.
Symmetric semigroups ($r = 0$) are excluded from $\mathcal{A}_{total}$: $r = 0$ forces every $r_i = 0$, hence $\rho = 0$, hence w_μ is not an atom.

9.2 $f + m$ atom: iterated descent, $\rho(S) > 0$

As long as $V(S) \subseteq S$ we can keep descending:

$$S_\rho \;\to\; D(S)_{\rho+1} \;\to\; D^2(S)_{\rho+2} \;\to\; \dots$$

At every step $f + m$ is the new atom w_μ (so $\mathcal{A}_{total}$ remains applicable), and ρ increases by 1. The chain terminates exactly when either $V(S) \not\subseteq S$ at the current S (so $\mathcal{D}_{total}$ no longer applies) or $f < m$ (so $V(S)$ is empty/degenerate).

10 Code

```
fn test_up_downs(s: &Semigroup) {
    if s.m == 1 {
        return;
    }
    for i in 1..s.m {
        let w = s.apery_set[i];

        // Descent: w-m is a gap; adding it lifts the membership of w-m.
        if w > s.f && s.f > s.m {
            assert!(!s.element(w - s.m));
            assert!(w <= s.f + s.m);
            assert!(
                w == s.f + s.m || s.is_reflected_gap(w - s.m),
                "reflected {w} at {i}"
            );
            let down = s.toggle(w - s.m);
            assert_eq!(down.g + 1, s.g);
            if w == s.f + s.m && s.v_in_s() {
                assert_eq!(down.r + 2, s.r + s.m, "r change w==f+m for i={i}");
                assert_eq!(down.sigma + s.m, s.sigma + 1, "sigma");
            }
            if w < s.f + s.m {
                assert_eq!(down.r + 2, s.r, "r change w<f+m for i={i}");
                assert_eq!(down.sigma, s.sigma + 1, "sigma");
            }
        }

        // Ascent: w must already be a minimal generator.
```

```
        if s.gen_set.contains(&w) {
            let up = s.toggle(w);
            if w < s.f && w + s.m > s.f {
                assert_eq!(
                    up.g,
                    s.g + 1,
                    "removing atom generates gap for {i} and {w}"
                );
                assert!(up.apery_set.contains(&(w + s.m)));
                assert_eq!(up.apery_set[i], w + s.m);
                assert_eq!(up.r, s.r + 2, "r change w<f for i={i}");
                assert_eq!(up.sigma + 1, s.sigma);
            }
            if w == s.f + s.m {
                let rho = s.rho();
                assert_eq!(
                    up.g,
                    s.g + rho,
                    "removing atom generates rho gap for {i} and {w}"
                );
                assert!(up.apery_set.contains(&(w + rho * s.m)));
                assert_eq!(up.apery_set[i], w + rho * s.m);
                assert_eq!(up.r + rho * (s.m - 2), s.r, "r change w==f+m for i={i}");
                assert_eq!(
                    up.sigma,
                    rho * (s.m - 1) + s.sigma,
                    "sigma change w==f+m for i={i}"
                );
            }
        }
    }
}
```

11 Claude's review notes

What this writeup is about, in plain words

You have a numerical semigroup S with multiplicity m and Frobenius number f. Its Apéry set has one element w_i per residue class mod m, and the largest of these is $w_\mu = f + m$ where $\mu = f \bmod m$. The writeup studies four local *moves* that turn S into a nearby semigroup:

$\mathcal{D}_{flip}$ **(descent, flip):** Pick an Apéry element w_i that lies in the open interval $(f, f+m)$ with $i \neq \mu$. Add $w_i - m$ to S. This fills in a reflected gap and lowers w_i by m.

$\mathcal{D}_{total}$ **(descent, total):** When the whole interval $V(S) = \{f-m+1, \dots, f-1\}$ already lies in S, add f itself. This collapses the Apéry structure dramatically: the new Frobenius drops below $f - m$, many Apéry elements get re-indexed at once, and σ decreases by $m - 1$.

$\mathcal{A}_{flip}$ **(ascent, flip):** The exact inverse of $\mathcal{D}_{flip}$: remove a generator $w_i \in V(S)$ with $i \neq \mu$.

$\mathcal{A}_{total}$ **(ascent, total):** The harder one. Remove $w_\mu = f + m$ as a generator. Because w_μ is the only way to reach the next few rungs of its residue ladder, you actually lose ρ elements in one move, where ρ is the smallest r_i over $i \neq \mu$.

The book-keeping (how g, σ, r change in each case) is mostly mechanical. What is interesting is whether you can *compose* these moves in a controlled way: take a descent followed by an ascent and ask when you get back to the same semigroup. That is the content of the "Correspondence" section. The punchline is: a single descent–ascent pair is a round trip only when the descent target has $\rho = 1$, i.e. the descent lifts a semigroup with w_μ *not* an atom to one with w_μ a uniquely-decodable atom. For higher ρ, you can iterate descents (provided $V(S) \subseteq S$ keeps holding), with each step increasing ρ by one and keeping $f + m$ an atom — until the chain hits a wall, namely a gap landing inside $V(S)$ or f dropping below m.
In short, this is a structural "hill-climbing" picture of the descent/ascent dynamics on the Kunz/Apéry side of the Wilf conjecture, with the four moves serving as elementary edges of an operation graph on numerical semigroups.

Status of the issues from the first pass

- **Case 4 definition.** Fixed: now $\langle A(S) \setminus \{w_\mu\} \rangle$, with a paragraph clarifying why the naive set-minus is the wrong object.
- $\rho > 0$ **hypothesis in Case 4.** Fixed: stated inline as "equivalently $\rho(S) > 0$", and the equivalence is proved in the Correspondence section.
- **Index inconsistency** $\mathcal{D}_{total}(i, \mu)$ **vs.** $\mathcal{A}_{total}(i, m)$**.** Fixed: the spurious second argument is gone everywhere; both families take only μ.
- **Codomain of** D**.** Fixed: D now lands in "numerical semigroups with multiplicity m" rather than $\mathbb{S}_\mu$. The old $\mathbb{S}_\mu$ has been renamed $\mathbb{S}_\mu^{\text{old}}$ and serves only as the domain of D.
- $r + \sigma = g$ **justification.** Fixed: a one-sentence reason (no $\{x, f - x\} \subset S_+$ by closure, pair-counting on $[0, f]$) appears in the Setup section.
- **"Viewed from below" equations.** Fixed: rewritten as ρ descents undoing one ascent, with the matching arithmetic.
- **Minor wording / typos.** Fixed: "necessesay / beeing / aways / The the / Semigroup", stray standalone S, range of L in RG, two Apéry definitions, ambiguous sum range, tautological "minimal generator" parenthetical, Wilf parenthetical.

Remaining issues (still open)

1. **No formal proofs.** The four change-formulas in Section UpDown are stated and informally justified ("proved by tracking the Apéry set"). The codebase property test `test_up_downs` checks them empirically over a wide fixture set, so they are very likely correct, but a reader of the writeup gets no algebraic derivation. Worth writing out at least one case (the $\sigma' = \sigma + \rho(m-1)$ case is the only one that surprises) as a small lemma.

2. $\rho(D(S)) = \rho(S) + 1$ **asserted, not proved.** The Correspondence section still says "the following properties are useful" without supplying arguments. The `s.r_i(i)` identity for Kunz coefficients is the natural proof tool, but the writeup doesn't reference it.

3. $V(D(S)) \subseteq D(S)$ **termination condition.** The iterated descent chain is described informally as "stops once there is a gap in V or $f < m$". There is presumably a clean characterisation of which semigroups admit a chain of length $\geq k$, but the writeup doesn't attempt one. Could be stated either combinatorially (in terms of the Apéry set) or via the Kunz matrix.

4. **Symmetry / direction conventions.** "Descent" *adds* elements to S; "ascent" *removes* them. This convention (descend = grow) is consistent with the codebase but is the reverse of what some readers might expect from the words. A one-line note up front ("descent shrinks the gap set; ascent enlarges it") would help.

5. **Wilf-aside is a teaser, not a result.** The line "we really should work out the change in e" is a TODO; no statement about how e moves under any of the four operations is given. If the writeup is meant to be self-contained even as an informal note, this is the most natural next section.

6. **Code listing is not commented on.** The verbatim Rust test in Section Code is given without text explaining which assertion corresponds to which case in Section UpDown. Adding case labels in comments (`// Case 1: D_flip`, etc.) would help the reader who wants to cross-check the math against the code.

7. **Where $\mathcal{D}_{flip}$ / $\mathcal{A}_{flip}$ live in the descent graph.** Section Correspondence treats only the *total* pair. Whether the flip pair admits any analogous round-trip discussion is not addressed. The inversion remark "$\mathcal{A}_{flip}$ inverts $\mathcal{D}_{flip}$" is a single sentence; the consequence (that the flip pair is always a round-trip on the appropriate domain) deserves a paragraph.

8. **Notation drift between r_i and $c(i, j)$.** Internally to the codebase, $r_i = c(i, \mu - i)$ where $c(\cdot, \cdot)$ is the Kunz coefficient. The writeup never introduces c, even though the Kunz interpretation makes some arguments (especially $\rho(D(S)) = \rho(S) + 1$) crisper.

Notation ↔ codebase

LaTeX	Code
m, f, μ	`s.m`, `s.f`, `s.mu`
g, σ, r	`s.g`, `s.sigma`, `s.r`
r_i, ρ	`s.r_i(i)`, `s.rho()`
w_i, $\mathrm{Ap}(S)$	`s.apery_set[i]`, `s.apery_set`
$\mathrm{RG}(S)$	`s.blob()`
$V(S) \subseteq S$	`s.v_in_s()`
atom	element of `s.gen_set`
$S^{\downarrow}$ (descent)	`s.toggle(w - s.m)`
$S^{\uparrow}$ (ascent)	`s.toggle(w)`
t	`s.t` (Rust) / `s.type_t` (JS)